

DATA 620 - Week 2 Part 2 Assignment

Centrality Measures

Using Centrality Measures to Study Artist Influence in the Spotify Feature Collaboration Network

Group Members: *Crystal Quezada, Nana Kwasi Danquah, and Muhammad Suffyan Khan*

1. Introduction

Centrality measures are useful in social network analysis because they help us identify which nodes are more connected, visible, or influential within a network. In a collaboration network, centrality can show which artists act as major collaborators, hubs, or bridges between groups.

For this assignment, we selected the Spotify Artist Feature Collaboration Network from Kaggle. The dataset represents feature and collaboration relationships between Spotify artists. Each artist is treated as a node, and a connection between two artists means they have appeared together through a collaboration relationship.

This dataset fits the assignment well because it has a clear network structure and also includes artist-level information that can be used for categorical comparison. For our main categorical variable, we would use music genre group. This allows us to ask whether artists from genres such as Hip Hop/Rap, Pop, Rock/Metal, Electronic/Dance, Latin, and R&B/Soul tend to be more central in the collaboration network.

2. Proposed Data Source

The proposed data source is the Spotify Artist Feature Collaboration Network, a publicly available dataset on Kaggle. The dataset contains two main CSV files: `nodes.csv` and `edges.csv`.

Based on the files reviewed for this assignment, `nodes.csv` contains 156,422 artist records and `edges.csv` contains 300,386 collaboration pairs.

Node file: `nodes.csv`

- `spotify_id`: unique Spotify artist identifier
- `name`: artist name
- `followers`: number of Spotify followers
- `popularity`: Spotify popularity score
- `genres`: list of genres associated with the artist
- `chart_hits`: chart appearance information

Edge file: `edges.csv`

- `id_0`: Spotify ID of the first artist in the collaboration pair
- `id_1`: Spotify ID of the second artist in the collaboration pair

In this network, nodes represent Spotify artists and edges represent feature or collaboration relationships between artists. We would treat the graph as undirected because a collaboration connects both artists to each other.

3. Categorical Variable for Node Comparison

The main categorical variable we would use is genre group. The original genres column contains detailed genre labels, and many artists have more than one genre listed. To make the comparison easier to interpret, we would create a simplified genre_group variable from the detailed genre labels.

For example, artists with labels such as “german hip hop,” “swedish hip hop,” “rap,” “trap,” or “drill” could be grouped into a broader Hip Hop/Rap category. Artists with labels such as “swedish pop” or “ukrainian pop” could be grouped into a broader Pop category.

Possible genre groups include:

- Hip Hop/Rap
- Pop
- Rock/Metal
- Electronic/Dance
- Latin
- Country/Folk
- Jazz/Soul/R&B
- Reggae/Dancehall
- Indie/Alternative
- Other

A second possible categorical variable is chart-hit status. Artists with non-empty chart_hits values could be classified as charting artists, while artists with empty chart_hits values could be classified as non-charting artists. This would allow a separate comparison of whether artists with chart history tend to be more central in the collaboration network.

4. Why This Dataset Is Interesting

Music collaborations are a major part of modern streaming culture. Artists often gain visibility by collaborating with other artists, especially when those collaborators already have large audiences or strong positions in the music industry.

A highly central artist in this network may have collaborated with many different artists. That could suggest broader industry reach, stronger cross-genre appeal, or more exposure to different listener communities. The dataset also lets us compare collaboration patterns across genres. Some genres, especially Hip Hop/Rap and Pop, often use featured artists, while other genres may be less collaboration-heavy.

Because of this, the dataset is a strong choice for comparing degree centrality across categorical groups.

5. High-Level Plan for Loading and Preparing the Data

This assignment does not require us to actually load or analyze the data. The steps below describe how we would prepare the data in Python for a degree centrality comparison.

Step 1: Load the CSV files

We would start by loading nodes.csv and edges.csv using pandas.

```
import pandas as pd

nodes = pd.read_csv("nodes.csv")
edges = pd.read_csv("edges.csv")
```

Step 2: Inspect and clean the data

Next, we would check the number of rows, column names, missing values, and duplicate rows. We would also confirm that the artist IDs in the edge list match the Spotify IDs in the node file.

```
nodes.head()
edges.head()
nodes.shape
edges.shape
nodes.isna().sum()
edges.duplicated().sum()
```

Step 3: Create a genre group variable

The raw genres column would be converted into a simpler genre_group variable. This step is important because the original genre labels are very detailed and would create too many small categories for a clean comparison.

```
def assign_genre_group(genres):
    genres = str(genres).lower()
    if "hip hop" in genres or "rap" in genres or "trap" in genres or "drill" in genres:
        return "Hip Hop/Rap"
    elif "pop" in genres:
        return "Pop"
    elif "rock" in genres or "metal" in genres or "punk" in genres:
        return "Rock/Metal"
    elif "dance" in genres or "electronic" in genres or "house" in genres or "edm" in genres:
        return "Electronic/Dance"
    elif "latin" in genres or "reggaeton" in genres:
        return "Latin"
    elif "country" in genres or "folk" in genres:
        return "Country/Folk"
    elif "r&b" in genres or "soul" in genres or "jazz" in genres:
        return "Jazz/Soul/R&B"
    else:
        return "Other"

nodes["genre_group"] = nodes["genres"].apply(assign_genre_group)
```

Step 4: Build the collaboration graph

The edge list would then be converted into a NetworkX graph. Since collaborations are mutual relationships, we would use an undirected graph.

```
import networkx as nx

G = nx.from_pandas_edgelist(
    edges,
    source="id_0",
    target="id_1",
    create_using=nx.Graph()
)
```

Step 5: Add node attributes

After building the graph, the artist metadata would be added as node attributes. These attributes would include artist name, followers, popularity, genres, chart hits, and the simplified genre group.

```
node_attributes = nodes.set_index("spotify_id").to_dict("index")
nx.set_node_attributes(G, node_attributes)
```

Step 6: Calculate degree centrality

Degree centrality would measure how many direct collaboration connections each artist has relative to the size of the network. Artists with high degree centrality would be direct collaboration hubs.

```
degree_centrality = nx.degree_centrality(G)
```

Step 7: Compare degree centrality across genre groups

Finally, we would merge the centrality scores back into the node dataframe and compare average degree centrality across genre groups.

```
centrality_df = pd.DataFrame({
    "spotify_id": list(degree_centrality.keys()),
    "degree_centrality": list(degree_centrality.values())
})

centrality_df = centrality_df.merge(
    nodes[["spotify_id", "name", "genre_group", "popularity", "followers", "chart_hits"]],
    on="spotify_id",
    how="left"
)

genre_summary = centrality_df.groupby("genre_group")["degree_centrality"].mean()
```

6. Hypothetical Outcome to Predict

A possible outcome that could be predicted using degree centrality is artist visibility or collaboration influence. Artists with higher degree centrality have more direct collaboration connections, which may give them more exposure to other artists' audiences and make them more visible within the Spotify collaboration network.

One hypothesis is that artists in Hip Hop/Rap and Pop will have higher average degree centrality than artists in some other genre groups because those genres commonly use featured artists and cross-artist collaborations.

Another hypothesis is that artists with chart-hit records will have higher average degree centrality than non-charting artists because charting artists are more visible and may have more opportunities to collaborate.

A related outcome is popularity. Artists with higher degree centrality may also have higher popularity scores or follower counts because collaborations can help artists reach new listener communities.

7. Expected Centrality Comparison

The main comparison would be between categorical groups. The table below shows the types of comparisons we could make.

Categorical Group	Possible Centrality Question
Genre group	Do Hip Hop/Rap, Pop, Rock/Metal, Electronic/Dance, Latin, and other groups differ in average degree centrality?

Chart-hit status	Are charting artists more central than non-charting artists?
Popularity group	Do high-popularity artists have higher degree centrality than lower-popularity artists?
Follower group	Do artists with larger follower counts have more collaboration connections?

For this assignment, the most important comparison would be genre group because genre is a meaningful category for music collaboration patterns.

8. Why Degree Centrality Is Relevant

Degree centrality is relevant because it measures how many direct connections a node has. In the Spotify collaboration network, an artist with high degree centrality has collaborated with many other artists.

This measure could help identify:

- artists who act as collaboration hubs,
- genres where feature-based collaboration is more common,
- artists who may have higher exposure across listener communities, and
- differences in collaboration patterns across categorical artist groups.

Degree centrality is especially appropriate here because the predicted outcome focuses on visibility and collaboration influence. More direct collaboration ties can give an artist more chances to reach new audiences and build a stronger position in the music network.

9. Relationship to Project 1

This Week 2 Part 2 assignment is separate from Project 1, even though our group used the same Spotify Artist Feature Collaboration Network for Project 1. Project 1 required the full analysis in a Jupyter Notebook, including loading the dataset, building the graph, calculating degree centrality and eigenvector centrality, comparing centrality across genre groups, and using statistical tests.

This assignment focuses on the planning side: identifying the dataset, explaining the categorical variables, describing how the data could be loaded, and presenting a hypothetical outcome that could be predicted by comparing degree centrality across categorical groups.

10. Conclusion

The Spotify Artist Feature Collaboration Network is a strong dataset for analyzing centrality measures because it represents real collaboration relationships among artists. It includes artist nodes, collaboration edges, and node-level information such as genres, popularity, followers, and chart-hit information.

For this assignment, we would focus on comparing degree centrality across genre groups. A high-level analysis could help show whether certain genres, such as Hip Hop/Rap or Pop, tend to be more central in the collaboration network than others. This could provide insight into how collaboration patterns differ across music categories and how centrality may relate to visibility or influence in streaming music.

Although this assignment does not require actual analysis, the plan shows how the data could be loaded, converted into a NetworkX graph, enriched with categorical attributes, and used to compare degree centrality across artist groups.

11. Video Presentation

A short video presentation explaining this assignment and the Spotify Artist Feature Collaboration Network centrality measures plan is available here:

Video Link: [DATA 620 Week 2 Part 2 Centrality Measures Presentation](#)

12. References

Freyberg, J. (n.d.). Spotify Artist Feature Collaboration Network. Kaggle.

<https://www.kaggle.com/datasets/jfreyberg/spotify-artist-feature-collaboration-network>

South, T., Roughan, M., & Mitchell, L. (2020). Popularity and Centrality in Spotify Networks: Critical transitions in eigenvector centrality. arXiv. <https://arxiv.org/abs/2008.11428>